

# **Bihar Groundwater (CGWB) and Child Health Burden (HMIS): Official Data Integration, Spatial Machine Learning, and District Susceptibility Profiles**

Comprehensive technical report: methodology, quantitative results, figures, susceptibility outputs,  
FAQ.

*Artefacts: data/processed/, results/models/, results/figures/, results/reports/  
PDF: D:\Major Project\results\reports\Bihar\_Major\_Project\_Comprehensive\_Report.pdf*

## **Table of contents**

1. Purpose and claims
2. Complete project definition
3. Data sources and file locations
4. End-to-end pipeline (how to reproduce)
5. Variables: features, target, derived columns
6. Statistical and ML methodology
7. Quantitative results (metrics explained)
8. SHAP and feature importance
9. Missing data
10. Susceptibility profiles (Yes/No outputs)
11. Figures (all result plots)
12. Anticipated questions and answers (FAQ)
13. Limitations and ethical framing
14. Appendix: artefact checklist

## 1. Purpose and claims

This project builds a reproducible pipeline that merges official Central Ground Water Board (CGWB) district-year groundwater summaries, HMIS Bihar child-health indicators, and Census 2011 district population into a single district-year table. Machine learning (Ridge, Random Forest, XGBoost) is trained with GroupKFold by district so test folds never contain districts seen during training (spatial generalisation stress test). Outputs include cross-validated metrics, out-of-fold predictions, SHAP-based explanations, publication-style figures, and district-year susceptibility profiles that separate environmental stress (CGWB-relative) from observed HMIS burden tiers.

**Primary claim: transparent integration and honest evaluation under spatial blocking. We do not claim that groundwater chemistry alone forecasts HMIS burden with high accuracy on unseen districts.**

## 2. Complete project definition

Problem domain: ecological (district-year) association between environmental groundwater summaries and administrative health burden proxies in Bihar. Unit of analysis: one row per District x Year after inner merge of CGWB aggregates and HMIS panel.

Software stack: Python 3; pandas; scikit-learn; optional XGBoost; SHAP; matplotlib; joblib; openpyxl for HMIS/Census reads; pymupdf used elsewhere for PDF extraction workflows.

Core Python package: bihar\_health\_risk (config, dataio, preprocess, train, explain, viz, susceptibility). CLI entry: scripts/run\_pipeline.py or python -m bihar\_health\_risk.

### 3. Data sources and file locations

Groundwater: CGWB Bihar well extracts stored under data/raw\_groundwater/cgwb\_bihar\_wells\_YYYY.csv (built via scripts/extract\_cgwb\_bihar\_csvs.py). Aggregated to district-year medians/means and merged.

HMIS: district-year indicators from the official HMIS workbook (see repository path used in scripts/build\_official\_master\_dataset.py). Burden composite sums selected child-health indicators under a fixed facility rollup (Total category).

Census 2011: district population for Bihar used as denominator for burden per 100k (static across years until projected populations are added).

**Master training table: data/processed/official\_district\_year\_master.csv (built by scripts/build\_official\_master\_dataset.py).**

## 4. End-to-end pipeline (reproduction)

Typical order:

- (1) `python scripts/extract_cgwb_bihar_csvs.py`
- (2) `python scripts/build_official_master_dataset.py`
- (3) `python scripts/run_pipeline.py`

`run_pipeline.py` loads the master CSV, writes `missingness.csv`, trains models with spatial CV, saves `metrics.json`, `best_model.joblib`, `oof_predictions.csv`, `shap_importance.csv`, `intervention_priority_top15.csv`, figures under `results/figures/`, and district-level susceptibility CSV/JSON.

## 5. Variables

### 5.1 Features (X)

Groundwater / monitoring features:

- GW\_pH\_median
- GW\_EC\_uScm\_median
- GW\_HCO3\_median
- GW\_Cl\_median
- GW\_SO4\_median
- GW\_NO3\_median
- GW\_F\_mgL\_median
- GW\_TDS\_median
- n\_wells

### 5.2 Target (y)

Target\_log\_burden\_per100k =  $\log(1 + \text{HMIS\_burden\_per\_100k})$ , where HMIS\_burden\_per\_100k is the composite HMIS raw burden sum divided by Census 2011 district population times 100,000. Log stabilises skewed rates.

### 5.3 Other columns used in susceptibility / reporting

High\_risk\_tercile: tertiles of HMIS\_burden\_per\_100k when computable. HMIS\_vulnerability\_z: mean z-score of  $\log_{10}(\text{indicator counts})$  across selected HMIS indicators.

## 6. Methodology

Preprocessing: numeric coercion, median imputation inside sklearn pipelines, optional clipping of non-negative GW constituents. Ridge uses scaled features; tree models use unscaled imputed features.

Spatial cross-validation: GroupKFold grouped by District. Number of splits is capped by district count (requested 5 folds when possible). This prevents leaking geographic structure across train and test.

Models compared: Ridge regression; Random Forest regressor; XGBoost regressor if installed. Best model chosen by lowest mean validation RMSE across folds; final pipeline refit on all rows for deployment artefacts.

Explainability: SHAP TreeExplainer / LinearExplainer on transformed feature matrix; mean absolute SHAP exported to results/models/shap\_importance.csv and summary bar plot saved as results/figures/shap\_summary.png.

## 7. Quantitative results

### 7.1 Project framing (embedded in metrics.json)

Primary goal:

Reproducible fusion of official CGWB + HMIS + Census data, with spatially honest validation (held-out districts). HMIS log-burden is an external referent for associational checks-not a forecasting target.

Success criteria:

- Transparent pipeline and artifacts
- GroupKFold by district (no leakage across geography)
- Compare OOF error to naive mean baseline
- Report rank alignment (Spearman) for prioritisation narratives
- SHAP for exposure directionality (hypothesis generation)

Why fold R2 can be negative:

Per-fold  $R^2$  uses small held-out district sets; predicting unseen districts with groundwater-only features is intentionally strict. Pooled OOF metrics and baseline comparison are clearer summaries for this study design.

### 7.2 Dataset scale

Districts: 33

District-year rows used after target handling: 63

Spatial CV folds: 5

Selected best model (by mean CV RMSE): rf

### 7.3 Pooled out-of-fold metrics (best model)

These summarise all rows using out-of-fold predictions from the selected model (RF in latest run).

- RMSE (pooled OOF): 1.5023319690286854
- R2 (pooled OOF vs global mean of y): -0.17884085314566533
- Spearman rho (rank correlation): -0.007680491551459294
- Spearman two-sided p-value: 0.9523610355161805
- Naive baseline RMSE (predict global mean y): 1.3836884631804012
- RMSE improvement vs naive (%): -8.574437744142397

Interpretation: negative improvement vs naive means the model's OOF RMSE is worse than always predicting the training-set mean of the target. That is consistent with weak marginal predictive signal from groundwater-only features under strict spatial holdout; it is reported explicitly rather than hidden.

### 7.4 Fold-averaged CV by model

Model ridge: mean RMSE=1.4963998402510368, mean MAE=1.229140521242112, mean fold R2=-0.4241229627750337

Model rf: mean RMSE=1.4941762017507663, mean MAE=1.2516691459523668, mean fold R2=-0.45033695903166693

Model xgb: mean RMSE=1.6698328862069123, mean MAE=1.4010916554698924, mean fold R2=-0.8016915837829967

## 8. SHAP importance (mean absolute SHAP)

GW\_HCO3\_median: 0.25722959218502567

GW\_Cl\_median: 0.1435313207938144

GW\_NO3\_median: 0.1418523430761691

GW\_EC\_uScm\_median: 0.13938390627100586

GW\_TDS\_median: 0.13827194656666583

GW\_pH\_median: 0.08430025011278264

GW\_F\_mgL\_median: 0.07663610437378232

n\_wells: 0.0700781738607569

GW\_SO4\_median: 0.06100970048786341

Reading guide: larger mean |SHAP| means the model relied more on that feature when explaining deviations from baseline prediction on the training distribution. This is not causal attribution.

## 9. Missingness (training columns)

GW\_TDS\_median: missing\_n=16, missing\_pct=25.40%

GW\_pH\_median: missing\_n=0, missing\_pct=0.00%

GW\_EC\_uScm\_median: missing\_n=0, missing\_pct=0.00%

GW\_HCO3\_median: missing\_n=0, missing\_pct=0.00%

GW\_SO4\_median: missing\_n=0, missing\_pct=0.00%

GW\_Cl\_median: missing\_n=0, missing\_pct=0.00%

GW\_NO3\_median: missing\_n=0, missing\_pct=0.00%

GW\_F\_mgL\_median: missing\_n=0, missing\_pct=0.00%

n\_wells: missing\_n=0, missing\_pct=0.00%

Target\_log\_burden\_per100k: missing\_n=0, missing\_pct=0.00%

Note: GW\_TDS\_median had notable missing share in the latest run; trees and imputers still run but interpretation should acknowledge reduced information on TDS for many rows.

## 10. Susceptibility profiles

File: results/models/district\_susceptibility\_profile.csv

Definitions JSON: results/models/susceptibility\_definitions.json

Yes if district-year is in top tertile of Environmental\_exposure\_index (mean z-score vs all district-years on CGWB features + log wells).

Yes if High\_risk\_tercile == high from HMIS burden per 100k (when column exists).

### 10.1 Summary counts from latest CSV

Rows (district-years): 63

Environmental high-stress Yes: 21 (33.3%)

HMIS observed high burden Yes: 21 (33.3%)

Both Yes (stress + high HMIS tertile): 6

Environmental Yes only (HMIS not high): 15

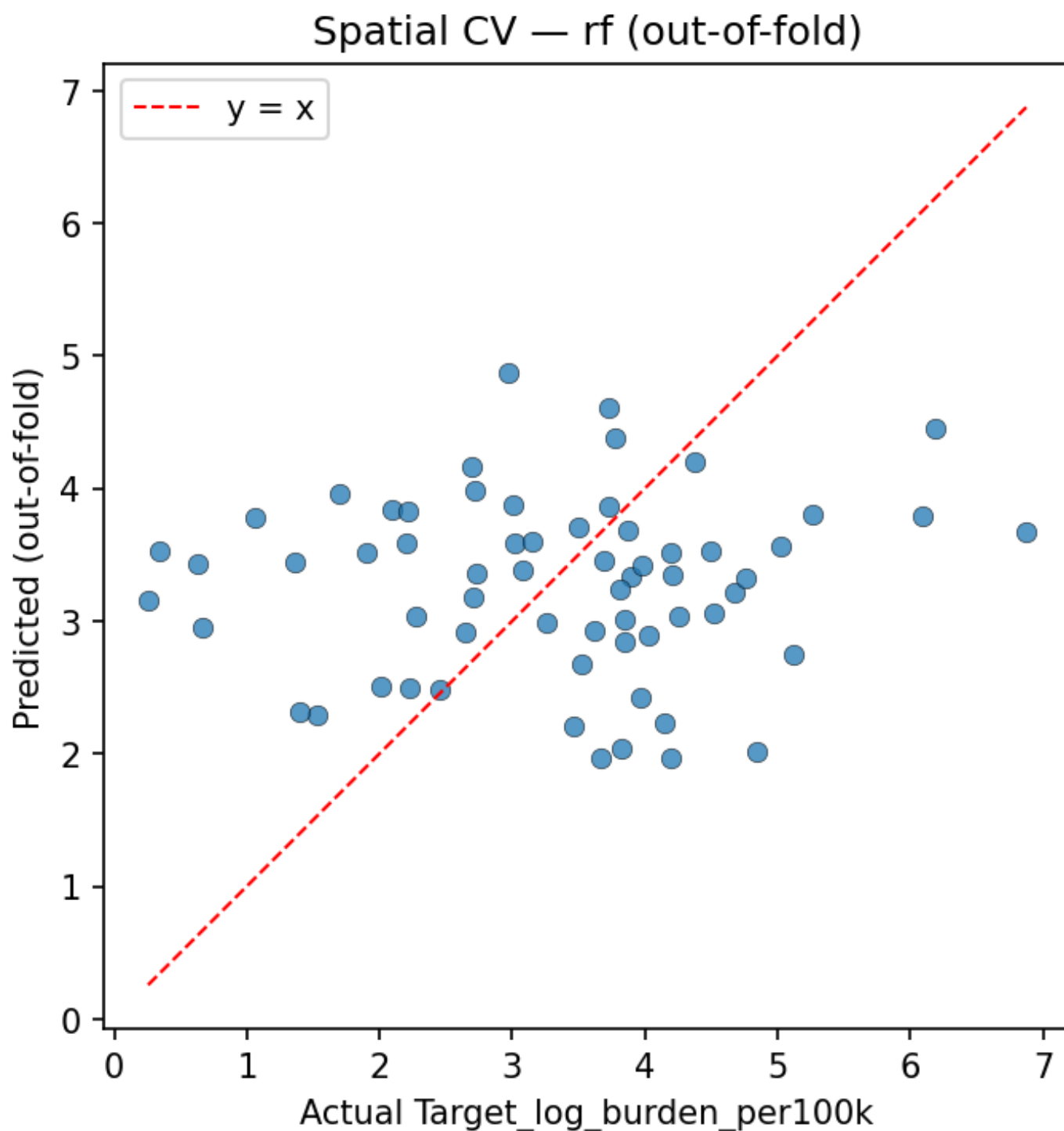
HMIS high only (environmental not top tertile): 15

### 10.2 HMIS indicators in composite

- Childhood Diseases - Severe Acute Malnutrition (SAM)
- Childhood Diseases - Diarrhoea
- Childhood Diseases - Pneumonia
- Childhood Diseases - Measles
- Number of Infant Deaths (1 -12 months) due to Diarrhoea
- Number of Infant Deaths (1 -12 months) due to Pneumonia
- Number of Child Deaths (1 -5 years) due to Diarrhoea
- Number of Child Deaths (1 -5 years) due to Pneumonia
- Number of cases of Infant deaths within 24 hrs of birth
- Number of still births

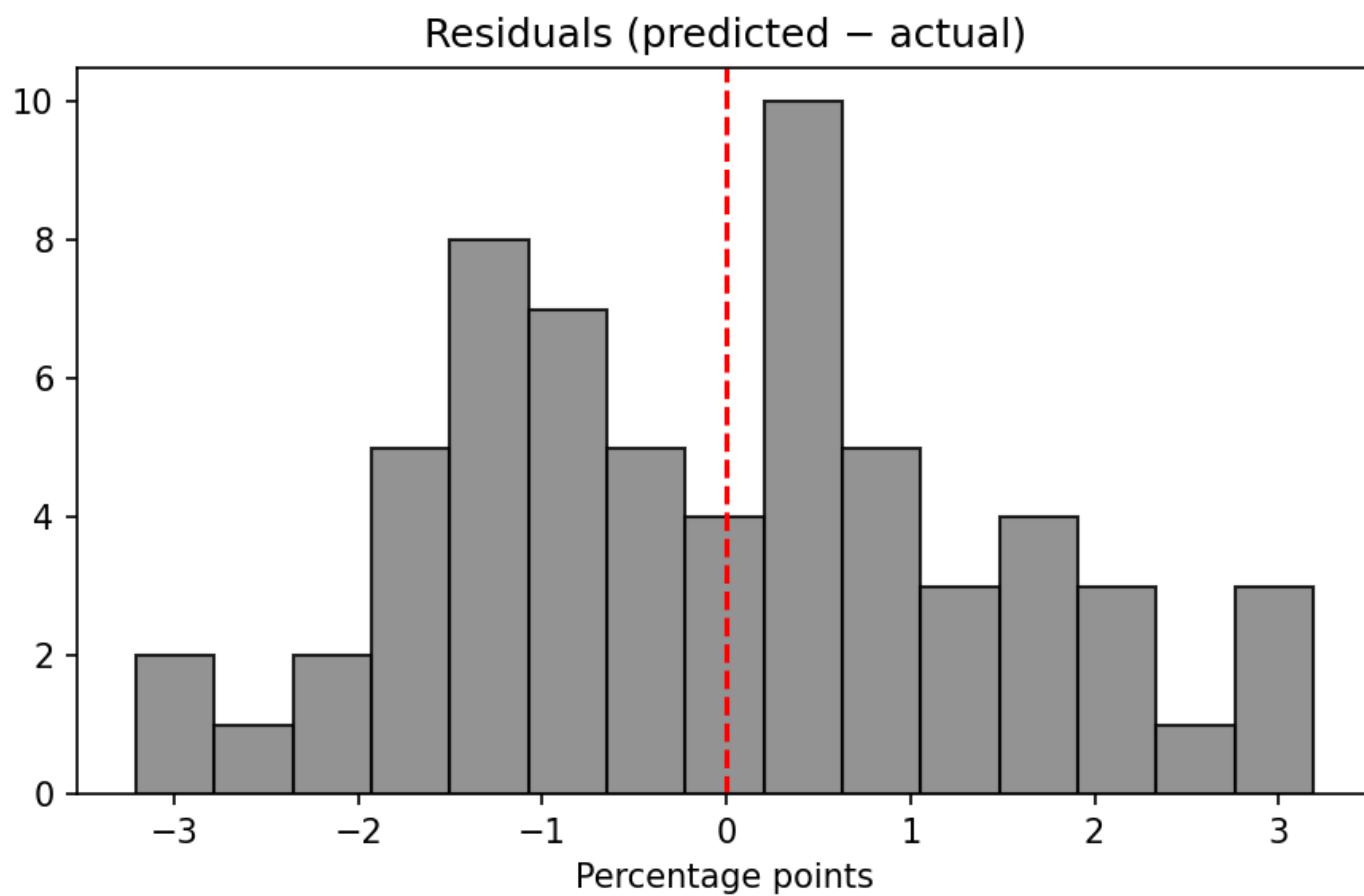
## 11. Figure: actual\_vs\_predicted.png

Figure: Actual vs predicted (spatial CV, out-of-fold). Each point is one district-year. x-axis: observed Target\_log\_burden\_per100k; y-axis: model prediction when that row was in a held-out district fold. Scatter around the diagonal reflects weak out-of-sample skill.



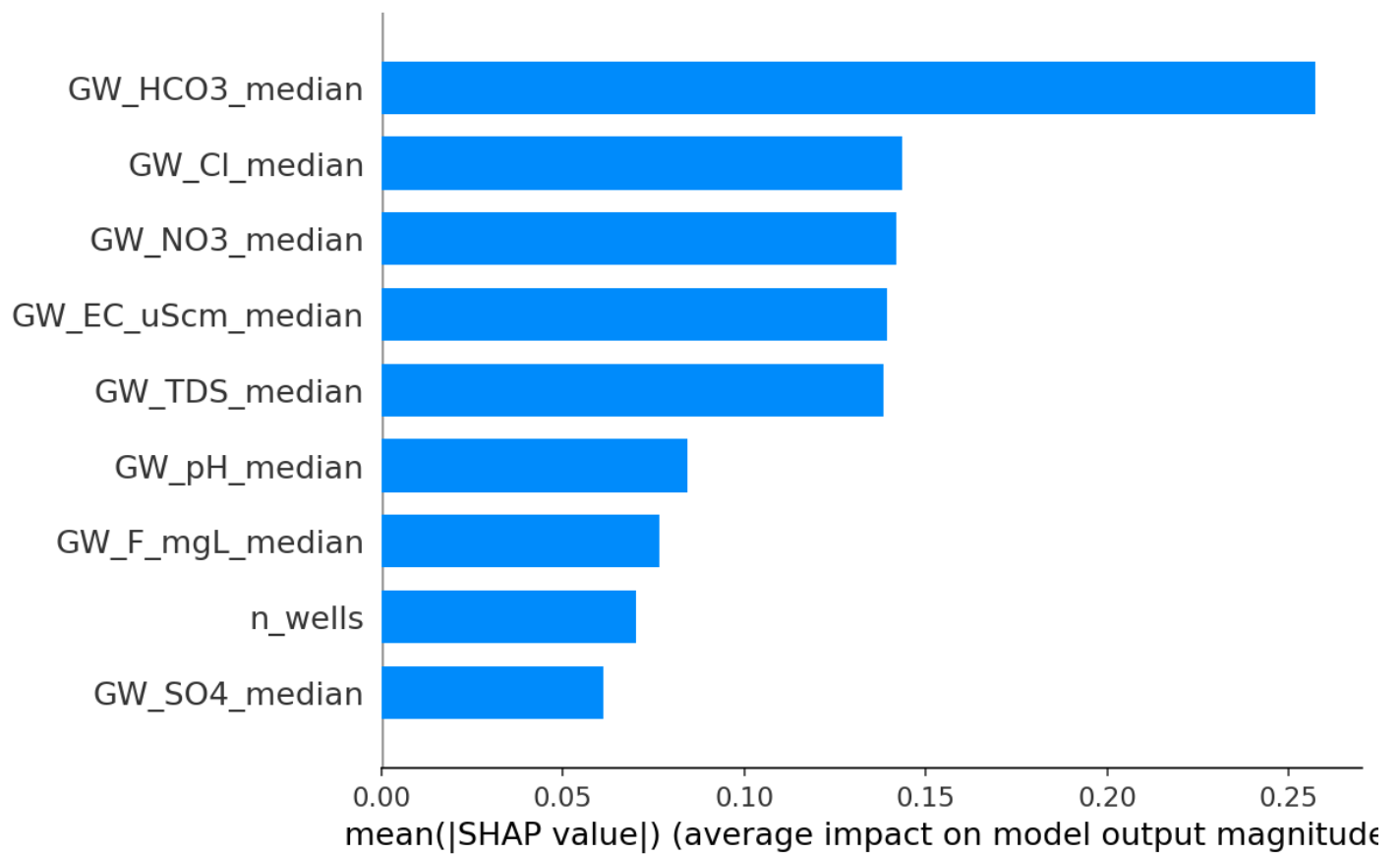
## 11. Figure: residuals.png

Figure: Residuals vs predicted values for OOF predictions. Checks systematic bias; funnel shapes may indicate heteroscedasticity.



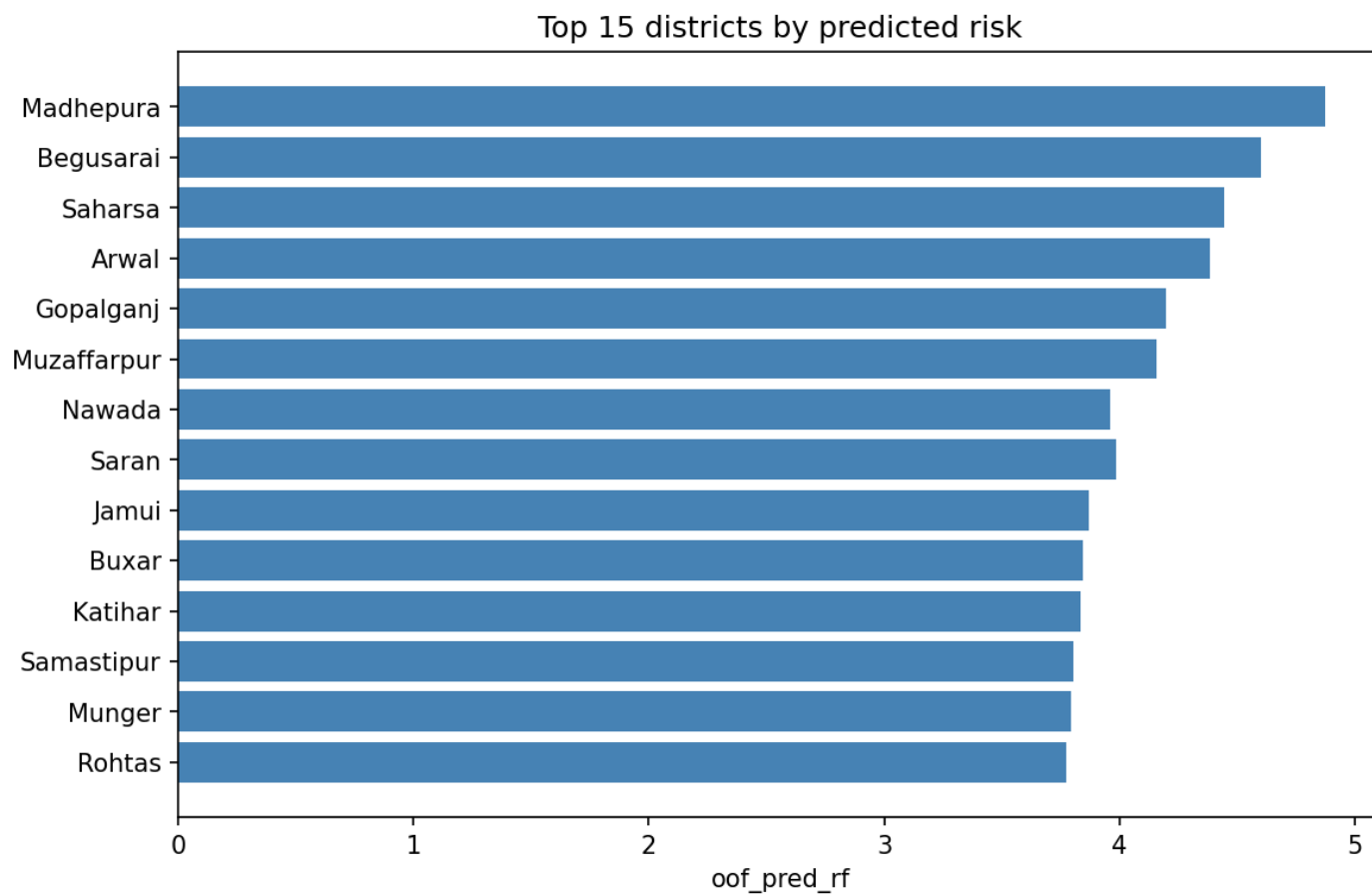
## 11. Figure: shap\_summary.png

Figure: SHAP summary (bar). Mean absolute SHAP value per groundwater/monitoring feature for the chosen model class.



## 11. Figure: top\_districts\_pred.png

Figure: Districts with highest out-of-fold predicted target under the selected model. Useful for relative ranking discussion; not a clinical severity certificate.



## 12. FAQ: anticipated examination questions

### **Q: What is the research question?**

A: Whether official groundwater summaries co-move with district-year HMIS child-health burden proxies when evaluated honestly at district scale, and whether we can produce transparent prioritisation artefacts from official data.

### **Q: Why district-year units?**

A: CGWB reporting and HMIS reporting align naturally at district administrative resolution for Bihar; year allows temporal variation where both sides exist.

### **Q: Why log burden target?**

A: Raw counts scale with population; rates per 100k reduce population dominance; log1p reduces skew from heavy-tailed rates.

### **Q: What is spatial GroupKFold?**

A: Cross-validation where entire districts are held out together so the model cannot memorise district-specific patterns that do not generalise geographically.

### **Q: Why is mean fold R2 negative?**

A: Small held-out sets and weak transferable signal make predictions worse than predicting the fold mean of y in some folds; averaging negative fold R2 stays negative.

### **Q: What is pooled OOF R2?**

A: R2 computed once using all out-of-fold predictions together vs the global mean of y; often smoother than averaging per-fold R2 but can still be negative.

### **Q: Why compare to naive mean RMSE?**

A: If a model cannot beat predicting the dataset mean under spatial holdout, it signals thin predictive marginal signal from X for that validation regime.

### **Q: Is Spearman rho near zero a failure?**

A: It indicates no meaningful rank alignment between OOF predictions and observed burden in this panel; consistent with weak GW-only association under spatial blocking.

### **Q: What does environmental susceptibility Yes mean?**

A: Top tertile of cohort-relative groundwater/monitoring stress index (z-score blend). Not a diagnosis.

### **Q: What does HMIS observed high burden Yes mean?**

A: Top tertile of HMIS composite burden per 100k among rows in the merged table when tertiles exist.

### **Q: Can environmental Yes and HMIS No happen together?**

A: Yes. Groundwater stress can be high while HMIS composite tertile is not high for many reasons (other drivers dominate, reporting effects, timing mismatch, ecological fallacy).

### **Q: Does SHAP prove causality?**

A: No. SHAP explains model dependence on features within the trained associations; confounding remains.

### **Q: Why is TDS sometimes missing?**

A: Well-level extraction and aggregation can leave TDS absent for some district-years; pipeline imputes.

### **Q: Why Census 2011 population for all years?**

A: Current master uses static denominator as approximation; projected populations would refine rates.

### **Q: Is this project meaningless?**

A: No if framed correctly: it delivers reproducible official-data fusion, strict validation, and explicit reporting of weakness. It is not meaningful as a sole groundwater-to-illness oracle.

### **Q: What would strengthen conclusions?**

A: Add socio-environmental covariates; refine population over time; consider count models with offsets; expand indicators carefully with HMIS string verification.

**Q: How do I regenerate everything?**

A: Run scripts/run\_pipeline.py after ensuring official\_district\_year\_master.csv exists.

**Q: Where are district Yes/No outputs?**

A: results/models/district\_susceptibility\_profile.csv plus susceptibility\_definitions.json.

**Q: Which model was finally saved?**

A: best\_model.joblib contains the pipeline refit on all rows; model\_name recorded inside bundle.

**Q: What is ecological bias?**

A: District aggregates cannot support individual-level causal claims; associations may reflect confounders.

### **13. Limitations and ethical framing**

- Thin predictor set (groundwater only) vs multifactorial burden.
- Administrative data artefacts (HMIS completeness; CGWB sampling intensity).
- Static population denominator.
- Small effective sample for strict spatial extrapolation.
- Composite HMIS sum mixes pathways; interpretation is coarse.

Ethical stance: avoid stigmatising districts; use outputs for prioritisation dialogues and hypothesis generation tied to official programmes, not blame assignment.

## 14. Appendix: artefact checklist

- data/processed/official\_district\_year\_master.csv
- results/models/metrics.json
- results/models/best\_model.joblib
- results/models/oof\_predictions.csv
- results/models/shap\_importance.csv
- results/models/missingness.csv
- results/models/intervention\_priority\_top15.csv
- results/models/district\_susceptibility\_profile.csv
- results/models/susceptibility\_definitions.json
- results/figures/actual\_vs\_predicted.png
- results/figures/residuals.png
- results/figures/top\_districts\_pred.png
- results/figures/shap\_summary.png
- results/reports/Official\_ML\_Project\_Report.md
- results/reports/Project\_Changes\_Detailed\_Report.md
- results/reports/Bihar\_Major\_Project\_Comprehensive\_Report.pdf (this document)